SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application-Based Questions 3Questions)

Course Code:	ITA0609	**Course Title:**	Machine Learning
CO Assessed:	CO1 – Analyze (BL 3)	**Assessment:**	Assessment Tool 1 – Application Based Questions.
Weightage:	40% of CIA	**Total Marks:**	30 (3 Questions ×10 Mark)
CO1 Statement:	*To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)*		
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Section / Batch:	SLOT A	**Date:**	13.7.26

Code of Conduct

I (N.kishan kumar-192424404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

Annexure A – Application-Based Questions: Answers

Question 1 — Probability (Bayes' Theorem)

Given:

- Section A: 5 red, 3 blue, 2 black → Total = 10 cars
- Section B: 4 red, 6 blue, 5 black → Total = 15 cars
- Section C: 7 red, 2 blue, 6 black → Total = 15 cars
- Section B is twice as likely to be chosen as Section A or Section C

Step 1: Prior probabilities of choosing each section

Let $P(A) = P(C) = x$ and $P(B) = 2x$

$x + 2x + x = 1 \rightarrow 4x = 1 \rightarrow x = 1/4$

$P(A) = 1/4$, $P(B) = 1/2$, $P(C) = 1/4$

Step 2: Probability of drawing a blue car from each section

$P(\text{Blue} \mid A) = 3/10$

$P(\text{Blue} \mid B) = 6/15 = 2/5$

$P(\text{Blue} \mid C) = 2/15$

Step 3: Total probability of drawing a blue car

$P(\text{Blue}) = P(A)\cdot P(\text{Blue}|A) + P(B)\cdot P(\text{Blue}|B) + P(C)\cdot P(\text{Blue}|C)$

$P(\text{Blue}) = (1/4 \times 3/10) + (1/2 \times 2/5) + (1/4 \times 2/15)$

$P(\text{Blue}) = 3/40 + 1/5 + 1/30$

Taking LCM = 120: $P(\text{Blue}) = 9/120 + 24/120 + 4/120 = 37/120$

Step 4: Apply Bayes' Theorem to find P(C | Blue)

$P(C \mid \text{Blue}) = [P(C)\cdot P(\text{Blue}|C)] / P(\text{Blue})$

$P(C \mid \text{Blue}) = (1/4 \times 2/15) / (37/120) = (4/120) / (37/120) = 4/37$

Answer: $P(C \mid \text{Blue}) = 4/37 \approx 0.108$ (about 10.8%)

Question 2 — Training vs Validation Loss

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2

Epoch	Training Loss	Validation Loss
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which the model begins to overfit

The training loss decreases steadily throughout all 8 epochs. The validation loss decreases up to Epoch 3 (2.0) but then starts rising from Epoch 4 onward (2.1, 2.3, 2.6, 2.9, 3.2), even though training loss keeps falling.

Answer: The model begins to overfit at Epoch 4, since this is where validation loss starts increasing while training loss continues to decrease.

ii) Two effective techniques to reduce overfitting

- Dropout Regularization: Randomly deactivating a fraction of neurons during training prevents the model from becoming overly dependent on specific neurons/features, improving generalization.
- Early Stopping: Monitoring validation loss during training and stopping once it starts to increase (around Epoch 3–4 here) prevents the model from continuing to memorize training data.

(Other acceptable techniques: L1/L2 regularization, data augmentation, reducing model complexity, batch normalization, k-fold cross-validation.)

iii) How overfitting affects the model's generalization ability

Overfitting occurs when a model learns not only the underlying patterns in the training data but also its noise and random fluctuations. As a result, the model achieves very low training loss but performs poorly on unseen/validation data, as reflected by the rising validation loss. This means the model fails to generalize — it memorizes training examples instead of learning patterns that transfer to new, unseen data, making it unreliable for real-world use.

Question 3 — Training vs Validation Accuracy

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

i) Epoch at which the model starts overfitting

Training accuracy increases continuously from 60% (Epoch 1) to 92% (Epoch 8). Validation accuracy increases up to Epoch 5 (73%), which is its highest point, and then starts declining from Epoch 6 onward (72%, 70%, 68%), even as training accuracy keeps rising.

Answer: The model starts overfitting from Epoch 6, since validation accuracy peaks at Epoch 5 (73%) and then drops in each subsequent epoch while training accuracy continues to rise.

ii) Why validation accuracy decreases despite training accuracy increasing

As training progresses beyond Epoch 5, the model becomes increasingly specialized to the specific examples, patterns, and noise present in the training set rather than learning generally applicable features. This makes the model's decision boundaries overly complex and tailored to training data. Consequently, while it keeps getting better at classifying training samples (higher training accuracy), its performance on new/unseen validation data worsens because the learned patterns do not generalize well a classic sign of overfitting and high variance.

iii) Two methods to improve validation performance

- Early Stopping: Stop training around the epoch where validation accuracy peaks (Epoch 5 here) instead of continuing to Epoch 8, preventing further overfitting.
- Data Augmentation / Regularization: Increasing the diversity of training data (augmentation) or adding regularization (dropout, L2 penalty) reduces the model's tendency to memorize training-specific details, helping validation accuracy remain stable or improve.